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A Finite Element Analysis of Second-Order Elliptic Boundary Value Problems

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Sincerely,
Arpan Saha

Abstract

In this project, we consider second order elliptic boundary value problems in the Sobolev space framework. Utilizing the Lax-Milgram theorem, the existence and uniqueness of weak solutions are proved, providing a strong foundation for numerical analysis. A finite element method is implemented for elliptic equations involving Laplace operators with Dirichlet boundary conditions on both convex (square) and non-convex (L-shaped) domains. A priori error estimates are discussed and convergence behaviour is examined through MATLAB simulations. While optimal convergence rates are observed on the square domain, reduced convergence is observed on the L-shaped domain, reflecting the loss of regularity due to the re-entrant corner. This highlights the susceptibility of finite element analysis to the geometry of domains and suggests the need for adaptive mesh refinement in non-smooth domains. The sequence of this report is as follows: The basics of finite element methods, particularly Sobolev spaces, Poincaré Inequality, Lax-Milgram theorem are discussed in Chapter 1. Then, weak solutions of second order elliptic boundary value problems are discussed rigorously in Chapter 2. After explaining Galerkin approximation in chapter 3, we introduce the finite element method and error analysis in Chapter 4. In chapter 5, we discussed a model problem. Later, we performed MATLAB simulations in chapter 6 for Dirichlet boundary conditions. Finally, we give a conclusion and mention some possible extensions of this project in Chapter 7. The project is mainly based on the work of Dr. Carstensen, Dr. Alberty, and Dr. Funken,*"Remarks around 50 lines of Matlab: short finite element implementation. Numerical algorithms, 20(2), 117-137"*; Dr. Brenner's and Dr. Scott's book,*"The mathematical theory of finite element methods"*; and Prof. S. Kesavan's book,*"Topics in functional analysis and applications"*.

Contents

Chapter 1

Preliminaries & Introduction

This chapter focuses on the basic concepts required for performing rigorous implementation of finite element methods. In this part, we study Sobolev spaces, Trace theorem, Poincaré inequality and Lax-Milgram theorem in detail. This chapter is particularly based on Prof. S. Kesavan's book,*"Topics in functional analysis and applications"*.

1.1 Sobolev Space

Let $\Omega \subset \mathbb{R}^N$ be an open set and let Γ denote its boundary.

Definition 1. Let m be a positive integer and let $1 \leq p \leq \infty$. The Sobolev Space $W^{m,p}(\Omega)$ is defined by,

$$W^{m,p}(\Omega) = \{u \in L^p(\Omega) \mid D^\alpha u \in L^p(\Omega), \text{ for all } |\alpha| \leq m\}.$$

It is a vector space contained in $L^p(\Omega)$, endowed with the norm $||\cdot||_{m,p,\Omega}$, defined as,

$$||u||_{m,p,\Omega} = \left(\sum_{|\alpha| \leq m} ||D^\alpha u||^p_{L^p(\Omega)} \right)^{1/p},$$

if $1 \leq p < \infty$ and

$$||u||_{m,\infty,\Omega} = \max_{|\alpha| \leq m} ||D^\alpha u||_{L^\infty(\Omega)}.$$

When $p = 2$, we will denote it by $H^m(\Omega)$, with The corresponding norm $||\cdot||_{m,2,\Omega}$, written as, $||\cdot||_{m,\Omega}$. $W^{1,p}(\Omega)$ is a Banach Space. Here, we prove this for $p = 2$.

Theorem 1. *$(H^1(\Omega), ||\cdot||_{1,\Omega})$ is a Banach space.*

Proof. Let $\{f_k\}$ be a Cauchy sequence in $H^1(\Omega)$. So, for given $\epsilon > 0$ be arbitrary, $\exists\, n_0 \in \mathbb{N}$ such that

$$||f_k - f_l||_{H^1(\Omega)} < \epsilon \;\forall\, k, l \geq n_0.$$

Therefore,

$$||f_k - f_l||_{L^2(\Omega)} < \epsilon \;\&\; \left\| \frac{\partial f_k}{\partial x_i} - \frac{\partial f_l}{\partial x_i} \right\|_{L^2(\Omega)} < \epsilon, i \in \{1, 2, ..., N\}.$$

Since, $L^2(\Omega)$ is complete, $f_k \to f$, $\frac{\partial f_k}{\partial x_i} \to g$ in $L^2(\Omega)$. To prove, $g = \frac{\partial f}{\partial x_i}$.
Now, for $\phi \in \mathcal{D}(\Omega)$,

$$\int_\Omega f_k \frac{\partial \phi}{\partial x_i} = -\int_\Omega \frac{\partial f_k}{\partial x_i}\phi$$
$$\Rightarrow \lim_{k\to\infty}\int_\Omega f_k \frac{\partial \phi}{\partial x_i} = -\lim_{k\to\infty}\int_\Omega \frac{\partial f_k}{\partial x_i}\phi$$
$$\Rightarrow \int_\Omega f \frac{\partial \phi}{\partial x_i} = -\int_\Omega g\phi,$$

where the last equality follows by the Cauchy-Schwartz inequality. So, the space is complete with the given norm. Hence, it is a Banach Space. □

Definition 2. The closure of the subspace $\mathcal{D}(\Omega)$ in $W^{m,p}(\Omega)$ is the closed subspace $W_0^{m,p}(\Omega)$. When $p = 2$, we denote this subspace by $H_0^m(\Omega)$.

1.2 Integration by Parts and Green's Formulas

Theorem 2 (Gauss-Green Theorem)**.** *Let U be an open set and $\nu = (\nu^1, \dots, \nu^n)$ denote the outward unit normal to ∂U.*

1. *Let $u \in C^1(\overline{U})$. Then, we have,*

$$\int_U u_{x_i}\,dx = \int_{\partial U} u\nu^i\,dS \quad (i = 1, \dots, n).$$

2. *Also,*

$$\int_U div\,\boldsymbol{u}\,dx = \int_{\partial U} \boldsymbol{u}\cdot\nu\,dS, \;\; \forall\,\boldsymbol{u} \in C^1(\overline{U}.\mathbb{R}^n)$$

Theorem 3 (Integration by parts formula)**.** *Let $u, v \in C^1(\overline{U})$. Then,*

$$\int_U u_{x_i} v\,dx = -\int_U u v_{x_i}\,dx + \int_{\partial U} uv\nu^i\,dS \quad \textit{where, } i = 1, \dots, n. \tag{1.1}$$

Theorem 4 (Green's formulas)**.** *Let $u, v \in C^2(\overline{U})$. Then,*

1. $\int_U \Delta u\,dx = \int_{\partial U} \frac{\partial u}{\partial \nu}\,dS.$
2. $\int_U Dv \cdot Du\,dx = -\int_U u\Delta v\,dx + \int_{\partial U} \frac{\partial v}{\partial \nu} u\,dS.$
3. $\int_U (u\Delta v - v\Delta u)\,dx = \int_{\partial U}\left(u\frac{\partial v}{\partial \nu} - v\frac{\partial u}{\partial \nu}\right)\,dS.$

1.3 Fractional Order Spaces and Trace Theorem

Suppose,

$$Q^+ = \{x \in \mathbb{R}^N \mid |x'| < 1, 0 < x_N < 1\}$$

where $|\cdot|$ stands for the usual Euclidean norm of the vector $x' = (x_1, \dots, x_{N-1}) \in \mathbb{R}^{N-1}$. The method of reflection gives an extension operator from $W^{1,p}(Q^+)$ into $W^{1,p}(Q)$ where

$$Q = \{x \in \mathbb{R}^N \mid |x'| < 1, |x_N| < 1\}.$$

Definition 3. We say that an open set $\Omega \subset \mathbb{R}^N$ is of class C^k, where $k \geq 1$ is an integer, if $\forall\, x \in \partial\Omega$, $\exists$ a neighbourhood U of x in $\mathbb{R}^N$ and a map $T : Q \to U$ such that,

(i) T is a bijection.

(ii) $T \in C^k(\overline{Q})$, $T^{-1} \in C^k(\overline{U})$.

(iii) $T(Q^+) = U \cap \Omega$, $T(Q_0) = U \cap \partial\Omega$.

where Q^+ and Q are as defined above and $Q_0 = \{x \in Q \mid x_N = 0\}$. We say that Ω is of class C^∞ if it is of class C^k for every positive integer k.

Sobolev Spaces $W^{s,p}(\Omega)$

Sobolev spaces $W^{s,p}(\Omega)$, $\Omega \subset \mathbb{R}^N$ can be defined in many ways.

Definition 4. If $1 \leq p < \infty$ and if $0 < \sigma < 1$, we can define,

$$W^{\sigma,p}(\Omega) = \left\{ u \in L^p(\Omega) \mid \frac{|u(x) - u(y)|}{|x - y|^{\frac{N}{p} + \sigma}} \in L^p(\Omega \times \Omega) \right\}.$$

Now, if $s = m + \sigma$, where m is an integer and $0 < \sigma < 1$, we define,

$$W^{s,p}(\Omega) = \{u \in W^{m,p}(\Omega) \mid D^\alpha u \in W^{\sigma,p}(\Omega) \text{ for all multi-indices } |\alpha| = m\}.$$

We denote by $W_0^{s,p}(\Omega)$ the closure of $\mathcal{D}(\Omega)$ in the space $W^{s,p}(\Omega)$. When $p = 2$, the spaces $W^{s,p}(\Omega)$ and $W_0^{s,p}(\Omega)$ are denoted by $H^s(\Omega)$ and $H_0^s(\Omega)$ respectively.

Definition 5. When $p = 2$ and $\Omega = \mathbb{R}^N$, we can define the spaces $H^s(\mathbb{R}^N)$ via the Fourier transform,

$$H^s(\mathbb{R}^N) = \{u \in L^2(\mathbb{R}^N) \mid (1 + |\xi|^2)^{s/2} \hat{u}(\xi) \in L^2(\mathbb{R}^N)\},$$

with corresponding norm given by,

$$\|u\|_{H^s(\mathbb{R}^N)} = \left(\int_{\mathbb{R}^N} (1 + |\xi|^2)^s |\hat{u}(\xi)|^2 \, d\xi \right)^{1/2}.$$

Theorem 5 (Trace Theorem)**.** *Let $\Omega \subset \mathbb{R}^N$ be a bounded open set of class C^{m+1}. Let $\Gamma = \partial\Omega$. Then there exist maps $\gamma_0, \gamma_1, \ldots, \gamma_{m-1}$ from $H^m(\Omega)$ into $L^2(\Gamma)$ such that,*

(i) If $v \in H^m(\Omega)$ is sufficiently smooth, then, $\gamma_0(v) = v|_\Gamma$, $\gamma_1(v) = \frac{\partial v}{\partial \nu}|_\Gamma$, $\gamma_{m-1}(v) = \frac{\partial^{m-1} v}{\partial \nu^{m-1}}|_\Gamma$, where ν is the outward unit normal on $\partial\Omega$.

(ii) The range of the map $(\gamma_0, \gamma_1, \ldots, \gamma_{m-1})$ is $\prod_{j=0}^{m-1} H^{m-j-\frac{1}{2}}(\partial\Omega)$.

(iii) The kernel of the map $(\gamma_0, \gamma_1, \ldots, \gamma_{m-1})$ is the space $H_0^m(\Omega)$.

1.4 Poincaré Inequality

Theorem 6. *Let Ω be a bounded open set in $\mathbb{R}^N$ and let $1 \leq p < \infty$. Then, there exists a constant $C = C(\Omega, p) > 0$ such that,*

$$\|u\|_{L^p(\Omega)} \leq C\|\nabla u\|_{L^p(\Omega)}, \forall\, u \in W_0^{1,p}(\Omega). \tag{1.2}$$

Proof. For $u \in \mathcal{D}(\Omega)$, let $\Omega \subset \{x = (x_1, x_2, ..., x_N) \in \mathbb{R}^N \mid -a < x_N < a\}$. We can write,

$$u(x) = \int_{-a}^{x_N} \frac{\partial u}{\partial x_N}(x', t)\, dt, \text{where } x' = (x_1, x_2, ..., x_{N-1}).$$

Applying Hölder's inequality, where $\frac{1}{p} + \frac{1}{q} = 1$,

$$|u(x)| \leq \left(\int_{-a}^{x_N} \left|\frac{\partial u}{\partial x_N}(x', t)\right|^p dt\right)^{1/p} |x_N + a|^{1/q}.$$

This implies,

$$|u(x)|^p \leq (2a)^{p/q} \int_{-a}^{a} \left|\frac{\partial u}{\partial x_N}(x', t)\right|^p dt.$$

Integrating with respect to x_N on $(-a, a)$,

$$\int_\Omega |u(x)|^p\, dx \leq (2a)^{p/q+1} \int_\Omega |\nabla u(x)|^p\, dx. \tag{1.3}$$

Using the fact that $\mathcal{D}(\Omega)$ is dense in $W_0^{1,p}(\Omega)$, we obtain, $\forall\, u \in W_0^{1,p}(\Omega)$,

$$\left(\int_\Omega |u|^p\, dx\right)^{1/p} \leq (2a) \left(\int_\Omega |\nabla u|^p\, dx\right)^{1/p}. \tag{1.4}$$

□

1.5 Lax-Milgram Theorem

Definition 6. (Continuous & H-elliptic bilinear form) Let H be a real Hilbert space. Suppose $a : H \times H \to \mathbb{R}$ be a bilinear form. It is defined as a continuous bilinear form if $\exists\ M > 0$ such that $|a(u, v)| \leq M||u||\ ||v||\ \ \forall\, u, v \in H$. It is defined to be H-elliptic if $\exists\ \alpha > 0$ such that $a(v, v) \geq \alpha||v||^2\ \ \forall\, v \in H$.

Example 1. Let $H = \mathbb{R}^N$ with usual Euclidean inner product. Let $A = (a_{ij}), 1 \leq i, j \leq N$. For $u = (u_1, ..., u_N)$, $v = (v_1, ..., v_N)$, we define the bilinear form, $a : \mathbb{R}^N \times \mathbb{R}^N \to \mathbb{R}$ as,

$$a(u, v) := v^T Au.$$

Then, we get,

$$|a(u, v)| \leq ||v||||Au|| \leq ||A||\ ||u||\ ||v||,$$

where the first inequality follows by Cauchy-Schwartz inequality. So, the bilinear form is continuous. Moreover, if A is symmetric and positive definite, we have,

$$|a(v, v)| = |v^T Av| \geq \alpha||v||^2,$$

where $\alpha > 0$ is the smallest eigenvalue of the matrix A. Then the bilinear form will be H-elliptic.

Theorem 7. *(Stampacchia)*
Let H be a Hilbert space and $a : H \times H \longrightarrow \mathbb{R}$ be a bilinear form which is continuous and H-elliptic. Let K be a closed convex set in H. Then given $f \in H$, $\exists!\, u \in K$ such that,

$$a(u, v - u) \geq \langle f, v - u \rangle \quad \forall\, v \in K. \tag{1.5}$$

Theorem 8. *(Lax-Milgram Theorem) Let H be a Hilbert space and $V \subseteq H$ be a closed subspace. Let $a(\cdot, \cdot)$ be a continuous, H-elliptic bilinear form on H. Then for any $f \in H$, $\exists!\, u \in V$ such that,*

$$a(u, v) = \langle f, v \rangle \quad \forall\, v \in V. \tag{1.6}$$

Moreover, the mapping $f \mapsto u = G(f)$ is a bounded linear map.

Proof. Since, V is subspace, it is convex set. Let $f \in H$. By Stampacchia theorem, $\exists!\, u \in V$ such that

$$a(u, v - u) \geq \langle f, v - u \rangle \quad \forall\, v \in V.$$

In that case, for any $w \in V$, set $v = w + u$. So, we get, $\forall\, w \in V$,

$$a(u, w) \geq \langle f, w \rangle.$$

Since V is a subspace, the above inequality also holds for $-w$. So, we have,

$$a(u, w) = \langle f, w \rangle \; \forall\, w \in V.$$

Now, if we set $w = u$, then

$$\alpha ||u||^2 \leq a(u, u) = \langle f, u \rangle \leq ||f||\, ||u||,$$

which implies,

$$||u|| \leq \frac{1}{\alpha} ||f||.$$

Thus, the mapping $f \mapsto u$ is a bounded linear map, and we are done. □

Chapter 2

Weak Solutions of Elliptic Boundary Value Problems

In this chapter, we consider different types of elliptic equations involving Laplace operators with various boundary conditions. This chapter is primarily based on Prof. S. Kesavan's book, *"Topics in functional analysis and applications"*.

2.1 The Homogeneous Dirichlet Problem

The Dirichlet Problem is defined on an open and bounded set $\Omega \subset \mathbb{R}^n$ with boundary $\Gamma = \partial\Omega$. We seek a function u such that:

$$\begin{cases} -\Delta u = f & \text{in } \Omega, \\ \quad u = 0 & \text{on } \Gamma, \end{cases} \tag{2.1}$$

where f is a given function.

Classical Solution

If $f \in C(\Omega)$, a **classical solution** is a function $u \in C^2(\bar{\Omega})$ that satisfies (2.1) pointwise.

Weak Formulation

If $u \in H_0^1(\Omega)$ satisfies,

$$\int_\Omega \nabla u \cdot \nabla \phi \, dx = \int_\Omega f\phi \, dx \quad \forall\, \phi \in H_0^1(\Omega), \tag{2.2}$$

then we say that, u is weak solution of (2.1).

Existence and Uniqueness

Theorem 9. *For a bounded open set Ω and $f \in L^2(\Omega)$, there exists a unique weak solution $u \in H_0^1(\Omega)$ of (2.1) satisfying (2.2).*

Proof. Here, $V = H_0^1(\Omega)$. Let $a(u,v) = \int_\Omega \nabla u.\nabla v \, dx$. So,

$$|a(u,v)| \leq \int_\Omega |\nabla u.\nabla v| \, dx \leq \|\nabla u\|_{L^2(\Omega)} \|\nabla v\|_{L^2(\Omega)},$$

where the last inequality follows by the Cauchy-Schwartz inequality. Hence, we can say that,

$$|a(u,v)| \le (\|\nabla u\|^2_{L^2(\Omega)} + \|u\|^2_{L^2(\Omega)})^{\frac{1}{2}} (\|\nabla v\|^2_{L^2(\Omega)} + \|v\|^2_{L^2(\Omega)})^{\frac{1}{2}} = \|u\|_{H^1_0(\Omega)} \|v\|_{H^1_0(\Omega)}.$$

Hence, a is a continuous bilinear form. Now, for $w \in H^1_0(\Omega)$,

$$a(w,w) = \int_\Omega |\nabla w|^2 dx = \|\nabla w\|^2_{L^2(\Omega)}.$$

So, we can write that,

$$\|w\|^2_{H^1_0(\Omega)} = \|w\|^2_{L^2(\Omega)} + \|\nabla w\|^2_{L^2(\Omega)} \le (C^2+1)\,\|\nabla w\|^2_{L^2(\Omega)}.$$

where the last inequality follows by Poincaré inequality. So, we have,

$$a(w,w) \ge \frac{1}{(C^2+1)} \|w\|^2_{H^1_0(\Omega)} \forall\, w \in H^1_0(\Omega).$$

Hence, a is a $H^1_0(\Omega)$-elliptic bilinear form and the result follows by Lax-Milgram theorem. □

2.2 The Inhomogeneous Dirichlet Problem

Consider f and g be given functions.

$$\begin{aligned} -\Delta u &= f \quad \text{in } \Omega, \\ u &= g \quad \text{on } \Gamma. \end{aligned} \tag{2.3}$$

We would like to look for a weak solution in $H^1(\Omega)$. Note that u is not in $H^1_0(\Omega)$, as $u = 0$ on Γ when in $H^1_0(\Omega)$, which is not the case here.
If $u \in H^1(\Omega)$, its trace belongs to $H^{1/2}(\Gamma)$. We assume that $g \in H^{1/2}(\Gamma)$. Thus, by the trace theorem, there exists $\tilde{u} \in H^1(\Omega)$ such that,

$$\tilde{u}|_\Gamma = \gamma_0(\tilde{u}) = g. \tag{2.4}$$

Now, define,

$$K = \tilde{u} + H^1_0(\Omega) = \{v \in H^1(\Omega) \mid v - \tilde{u} \in H^1_0(\Omega)\}. \tag{2.5}$$

We seek $u \in K$. Define a weak solution as $u \in K$ such that

$$\int_\Omega \nabla u \cdot \nabla v \, dx = \int_\Omega f v \, dx \quad \forall v \in H^1_0(\Omega). \tag{2.6}$$

For the existence of a weak solution u, set $u = \tilde{u} + w$ where $w \in H^1_0(\Omega)$. So, we seek $w \in H^1_0(\Omega)$ such that,

$$\int_\Omega \nabla w \cdot \nabla v \, dx = \int_\Omega f v \, dx - \int_\Omega \nabla \tilde{u} \cdot \nabla v \, dx \quad \forall\, v \in H^1_0(\Omega). \tag{2.7}$$

2.3 The Neumann Problem

Let ν be the unit outer normal on the boundary of a bounded open set $\Omega \subset \mathbb{R}^n$. We denote the exterior normal derivative on Γ of a smooth function u defined on Ω as $\frac{\partial u}{\partial \nu}$. So,

$$\frac{\partial u}{\partial \nu} = \nabla u \cdot \nu.$$

Let $f : \Omega \to \mathbb{R}$ is a given function. Then consider the following boundary value problem,

$$\begin{cases} -\Delta u + u = f \text{ in } \Omega, \\ \frac{\partial u}{\partial \nu} = 0 \text{ on } \Gamma. \end{cases} \tag{2.8}$$

If u is a classical solution of (2.8), then $u \in H^1(\Omega)$. Using Green's formula, $\forall\, v \in H^1(\Omega)$,

$$\int_\Omega \nabla u \cdot \nabla v \, dx + \int_\Omega uv dx = \int_\Omega fv \, dx. \tag{2.9}$$

If $f \in L^2(\Omega)$, then we define a weak solution of (2.8) as a function $u \in H^1(\Omega)$ satisfying (2.9) $\forall\, v \in H^1(\Omega)$. Let

$$a(u, v) = \int_\Omega \nabla u \cdot \nabla v \, dx + \int_\Omega uv \, dx.$$

This is a inner-product in $H^1(\Omega)$. Hence, we have symmetry, continuity, and $H^1(\Omega)$-ellipticity of the bilinear form, and the existence of a unique weak solution follows from the Lax-Milgram theorem. Also, we should mention that, if $u \in H^2(\Omega) \subset H^1(\Omega)$ is a weak solution, it satisfies (2.8) in the sense of distributions and the boundary conditions in the sense of trace. Moreover, if $u \in C^2(\overline{\Omega})$ and $f \in C(\overline{\Omega})$, then u will be a classical solution of (2.8).

2.4 The Mixed Problem

Another type of problem is the mixed boundary value problem,

$$\begin{cases} -\Delta u = f \text{ in } \Omega, \\ u = 0 \text{ on } \Gamma_1, \\ \frac{\partial u}{\partial \nu} = 0 \text{ on } \Gamma_2, \end{cases} \tag{2.10}$$

where $\Gamma = \Gamma_1 \cup \Gamma_2$ and $\Gamma_1 \cap \Gamma_2 = \emptyset$. Suppose, surface measure of Γ_1 is strictly positive. Then Poincaré's inequality is still available for the space,

$$V = \{v \in H^1(\Omega) \mid v|_{\Gamma_1} = 0\}.$$

.

So, we can apply the Lax-Milgram theorem for the problem: find $u \in V$ such that

$$\int_\Omega \nabla u \cdot \nabla v \, dx = \int_\Omega fv \, dx, \tag{2.11}$$

$\forall\, v \in V$. This is the weak formulation of the mixed problem (2.10) stated above.

Chapter 3

Galerkin Approximation

Suppose we have to prove the existence of a solution to the problem: find $u \in H_0^1(\Omega)$ such that

$$a(u, v) = \langle f, v \rangle \tag{3.1}$$

for every $v \in H_0^1(\Omega)$, where $\langle f, v \rangle = \int_\Omega fv\, dx$, $a(u, v) = \int_\Omega \nabla u . \nabla v \; dx \; \forall \; v \in H_0^1(\Omega), f \in L^2(\Omega)$. When a is not $H_0^1(\Omega)$-elliptic, we cannot use Lax-Milgram theorem. In that case, we may use the Galerkin method. The Galerkin Method is a method of approximation of solutions of abstract functional equations. It can be used to prove the existence of solutions to an equation by proving the convergence of the approximate solutions in a suitable topology. However, here, we have the coercivity of the bilinear form. This chapter is based on Prof. S. Kesavan's book, *"Topics in functional analysis and applications"*.

3.1 Existence of solutions in finite dimensional subspaces

Suitable basis

Since, $H_0^1(\Omega)$ is a separable Hilbert space, we have a countable orthonormal basis for it, which we denote by $\{e_1, e_2, ...\}$. For every positive integer m, let

$$V_m = \text{span}\{e_1, e_2, ..., e_m\}.$$

Existence of solutions

We now prove the existence of $u_m \in V_m$ such that

$$a(u_m, v_m) = \langle f, v_m \rangle \tag{3.2}$$

for every $v_m \in V_m$.

Matrix Formulation

Let $u_m \in V_m$ be such that

$$u_m = \sum_{j=1}^{m} c_j e_j,$$

for some scalars $c_1, c_2, ..., c_m$ in $\mathbb{R}$. Then, from (3.2),

$$c_1 a(e_1, v_m) + c_2 a(e_2, v_m) + ... + c_m a(e_m, v_m) = \langle f, v_m \rangle \; \forall \; v_m \in V_m.$$

So, we have the following system of equations,

$$\begin{aligned}
c_1 a(e_1, e_1) + c_2 a(e_2, e_1) + ... + c_m a(e_m, e_1) &= \langle f, e_1 \rangle, \\
c_1 a(e_1, e_2) + c_2 a(e_2, e_2) + ... + c_m a(e_m, e_2) &= \langle f, e_2 \rangle, \\
c_1 a(e_1, e_3) + c_2 a(e_2, e_3) + ... + c_m a(e_m, e_3) &= \langle f, e_3 \rangle, \\
&\vdots \\
c_1 a(e_1, e_m) + c_2 a(e_2, e_m) + ... + c_m a(e_m, e_m) &= \langle f, e_m \rangle.
\end{aligned}$$

Hence, we have the matrix form,

$$AU = F, \tag{3.3}$$

where,

$$A = \begin{bmatrix} a(e_1, e_1) & a(e_2, e_1) & a(e_3, e_1) & ... & a(e_m, e_1) \\ a(e_1, e_2) & a(e_2, e_2) & a(e_3, e_2) & ... & a(e_m, e_2) \\ \vdots & \vdots & \vdots & & \vdots \\ a(e_1, e_m) & a(e_2, e_m) & a(e_3, e_m) & ... & a(e_m, e_m) \end{bmatrix}, U = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_m \end{bmatrix}, \text{ and } F = \begin{bmatrix} \langle f, e_1 \rangle \\ \langle f, e_2 \rangle \\ \vdots \\ \langle f, e_m \rangle \end{bmatrix}.$$

By definition of the bilinear form a, the matrix A is positive definite. So, it is invertible and the system has a unique solution. So, u_m is the unique solution of (3.2).

3.2 Uniform boundedness of the approximate solutions

By Poincaré inequality,

$$||u_m||^2_{H^1_0(\Omega)} = \|\nabla u_m\|^2_{L^2(\Omega)} + \|u_m\|^2_{L^2(\Omega)} \leq (C^2 + 1)\|\nabla u_m\|^2_{L^2(\Omega)}$$

Now, by definition of bilinear form for homogeneous Dirichlet problem,

$$||u_m||^2_{H^1_0(\Omega)} \leq (C^2 + 1)\, a(u_m, u_m) = (C^2 + 1)\langle f, u_m \rangle.$$

Using Cauchy-Schwarz inequality, we get,

$$||u_m||^2_{H^1_0(\Omega)} \leq (C^2 + 1)||f||_{L^2(\Omega)}||u_m||_{L^2(\Omega)} \leq (C^2 + 1)||f||_{L^2(\Omega)}||u_m||_{H^1_0.(\Omega)}$$

Hence, we have,

$$||u_m||_{H^1_0(\Omega)} \leq (C^2 + 1)||f||_{L^2(\Omega)}. \tag{3.4}$$

Therefore, the approximate solutions are uniformly bounded.

3.3 Convergence of the approximate solutions

We know that every bounded sequence has a weakly convergent subsequence in a Hilbert space. Let u_{m_k} be that subsequence and $u \in H^1_0(\Omega)$ be that weak limit. Let $v \in H^1_0(\Omega)$. Consider,

$$v_m = \sum_{j=1}^{m} \langle v, e_j \rangle e_j.$$

Then, $v_m \to v$ as $m \to \infty$ in $H_0^1(\Omega)$. Again,

$$a(u_{m_k}, v_{m_k}) = \langle f, v_{m_k} \rangle. \tag{3.5}$$

So, now we have,

$$\begin{aligned}\big|a(u_{m_k}, v_{m_k}) - a(u,v)\big| &= \big|a(u_{m_k}, v_{m_k}) - a(u_{m_k}, v) + a(u_{m_k}, v) - a(u,v)\big| \\ &\leq \big|a(u_{m_k}, v_{m_k} - v)\big| + \big|a(u_{m_k}, v) - a(u,v)\big|.\end{aligned}$$

Using the continuity of the bilinear form $a(\cdot,\cdot)$, there exists $M > 0$ such that

$$|a(w,s)| \leq M \, \|w\|_{H_0^1(\Omega)} \, \|s\|_{H_0^1(\Omega)} \qquad \forall \, w, s \in H_0^1(\Omega).$$

Hence,

$$\big|a(u_{m_k}, v_{m_k} - v)\big| \leq M \, \|u_{m_k}\|_{H_0^1(\Omega)} \, \|v_{m_k} - v\|_{H_0^1(\Omega)}.$$

Since, the sequence $\{u_{m_k}\}$ is bounded, there exists $C > 0$ such that

$$\|u_{m_k}\|_{H_0^1(\Omega)} \leq C,$$

and consequently

$$\big|a(u_{m_k}, v_{m_k} - v)\big| \leq MC \, \|v_{m_k} - v\|_{H_0^1(\Omega)}.$$

Therefore, $\big|a(u_{m_k}, v_{m_k} - v)\big| \to 0$ as $k \to \infty$ since $v_m \to v$ as $m \to \infty$ in $H_0^1(\Omega)$. Now, for fixed $v \in H_0^1(\Omega)$, the map

$$T_v : H_0^1(\Omega) \to \mathbb{R}, \text{ defined by, } T_v(w) = a(w,v),$$

is a bounded linear functional. Since,

$$u_{m_k} \rightharpoonup u \qquad \text{in } H_0^1(\Omega),$$

it follows from the definition of weak convergence that

$$a(u_{m_k}, v) = T_v(u_{m_k}) \longrightarrow T_v(u) = a(u,v) \text{ as } k \to \infty.$$

Combining these above estimates, we obtain

$$a(u_{m_k}, v_{m_k}) \longrightarrow a(u,v).$$

On the other hand, since $v_{m_k} \to v$ strongly in $H_0^1(\Omega)$,

$$\langle f, v_{m_k} \rangle \longrightarrow \langle f, v \rangle \text{ as } k \to \infty.$$

So, Passing to the limit in (3.5), we get,

$$a(u,v) = \langle f, v \rangle.$$

Since $v \in H_0^1(\Omega)$ was arbitrary, we conclude that

$$a(u,v) = \langle f, v \rangle \qquad \forall \, v \in H_0^1(\Omega).$$

Therefore, u is the weak solution of (3.1).

Chapter 4

Introduction to the Finite Element Method

In this chapter, we will study the introduction to finite element analysis. After discussing the whole procedure in brief, we will discuss the formation of fnite dimensional subspaces in one dimension and two dimension. After that we will discuss the triangulation process, and the finite dimensional spaces. Lastly, we will derive the Céa's Lemma for error analysis. This chapter is mainly based on Prof. S. Kesavan's book,*"Topics in functional analysis and applications"* and Dr. Brenner's and Dr. Scott's book,*"The mathematical theory of finite element methods"*.

4.1 Procedure of the Finite Element Analysis

The finite element analysis of a physical problem consists of a systematic construction of finite-dimensional subspaces of various Sobolev spaces. The standard procedure consists of the following fundamental steps:

- **Step 1: Variational Formulation.** Given an elliptic boundary value problem, it is transformed into its weak form. The existence and uniqueness of the solution $u \in V$ are typically proven using the Lax-Milgram lemma.
- **Step 2: Discretization.** Once the space V, the bilinear form $a(\cdot,\cdot)$, and the functional $f \in V'$ are identified, a finite-dimensional subspace $V_h \subset V$ is constructed.
- **Step 3: Basis Selection and Matrix Assembly.** A basis for V_h is identified such that the resulting stiffness matrix A_h is as sparse as possible (i.e., $a_{ij}^h = 0$ for most entries). This sparsity is essential for computational efficiency, allowing for the storage of large-scale banded matrices. As $h \to 0$, V_h approaches V, leading to a very large dimension $n(h)$.
- **Step 4: Error Analysis.** First, study the interpolation error between V and V_h. Obtaining specific error inequalities usually requires the solution $v \in V$ to be sufficiently regular; for instance, if $V = H^1(\Omega)$, we typically require $v \in H^2(\Omega)$.

The formation of these subspaces and the specifics of the error analysis mentioned above will be discussed in detail in the following two subsections.

4.2 Formation of Finite Dimensional Subspaces

4.2.1 1-Dimensional Case

To illustrate the construction of V_h in the one-dimensional case, let $\{x_0, x_1, \ldots, x_n\}$ be a partition of $[0, 1]$, where $0 = x_0 < x_1 < \cdots < x_n = 1$. We define S as the linear space of functions $v : [0, 1] \to \mathbb{R}$ such that,

1. $v \in C([0, 1])$
2. $v|_{[x_{i-1}, x_i]}$ is a linear polynomial for $i = 1, \ldots, n$, and
3. $v(0) = v(1) = 0$.

Construction of Basis Functions

For each $i = 1, \ldots, n$, we define the basis functions $\phi_i : [0, 1] \to \mathbb{R}$ by,

$$\phi_i(x_j) = \delta_{ij} \tag{4.1}$$

where δ_{ij} is the Kronecker delta.

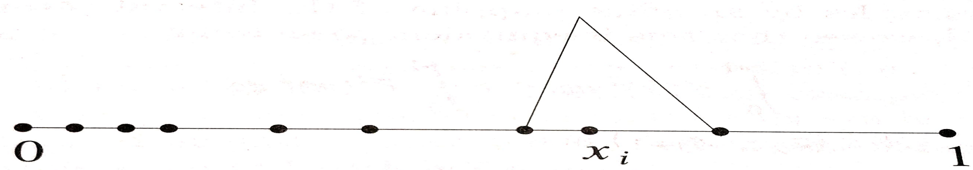

Figure 4.1: Piecewise linear basis function ϕ_i

$\{\phi_i\}$ is referred to as the **nodal basis**, and $\{v(x_i)\}$ are the **nodal values** of a function $v : [0, 1] \to \mathbb{R}$. Next, we will prove that the collection, $\{\phi_i\}$ is a basis for S.

Definition 7. (Interpolant). Given $v \in C([0, 1])$, the interpolant $v_I \in S$ of v is determined by

$$v_I := \sum_{i=1}^{n} v(x_i)\phi_i.$$

Lemma 1. $v \in S \Rightarrow v = v_I$.

Proof. v_I is linear on each $[x_{i-1}, x_i]$, $v(0) = 0$. Now, $v_I(0) = \sum_{i=1}^{n} v(x_i)\phi_i(0) = 0$. Hence, $v(0) = v_I(0)$. Again, $v(x_j) - v_I(x_j) = v(x_j) - \sum_{i=1}^{n} v(x_i)\phi_i(x_j) = v(x_j) - v(x_j) = 0$.
$\Rightarrow v(x_j) - v_I(x_j) = 0$ for $j = 1, \ldots, n$. Now, $(v - v_I)(x) = ax + b$ in each sub-interval, $a, b \in \mathbb{R}$. $(v - v_I)(0) = 0 \Rightarrow a \cdot 0 + b = 0 \Rightarrow b = 0$.
Now, putting another nodal point x_j, $ax_j = 0 \Rightarrow a = 0$, since $x_j \neq 0$. So, $\Rightarrow v = v_I = \sum_{i=1}^{n} v(x_i)\phi_i$. $\square$

Lemma 2. *The set* $\{\phi_i : 1 \leq i \leq n\}$ *is a basis for* S.

Proof. **Linear Independence:** $\sum_{i=1}^{n} c_i\phi_i(x_j) = 0$ implies $c_j = 0$ for each j, thus the functions are linearly independent.
Spanning: This follows directly from **Lemma 1**.
Hence, $\{\phi_i : 1 \leq i \leq n\}$ is a basis for S. $\square$

Expression of Finite Element Basis Functions

So, from the definition of basis functions, we get, by successive calculations,

$$\phi_i(x) = \begin{cases} \frac{x-x_{i-1}}{x_i-x_{i-1}} & \text{if } x \in [x_{i-1}, x_i], \\ \frac{x_{i+1}-x}{x_{i+1}-x_i} & \text{if } x \in [x_i, x_{i+1}], \\ 0 & \text{otherwise.} \end{cases} \tag{4.2}$$

4.2.2 2-Dimensional Case

In $\mathbb{R}^2$, the basic idea to construct the subspaces V_h is to partition the bounded polygonal domain $\Omega \subset \mathbb{R}^2$ into smaller polygons, typically triangles.

Triangulation and Mesh Size

A triangulation $\mathcal{F}_h$ of Ω is a partition of Ω into triangles such that each side is either part of the boundary Γ of Ω, or is a side of an adjacent triangle.

- **Admissibility:** We do not allow triangulations where a vertex of one triangle lies on the interior of the side of another (see Figure 4.2).

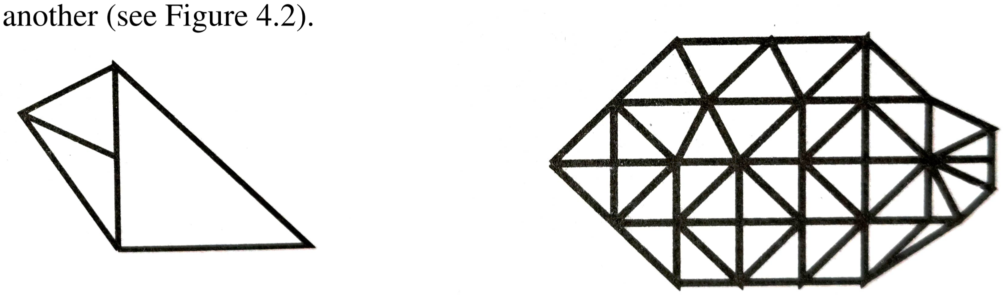

Figure 4.2: Process of triangulation (The picture in the left side does not satisfy the criterion for triangulation, but the picture in the right side is a correct example of triangulation)

- **Element Definition:** Each triangle K is referred to as an *element* of the triangulation.

- **Mesh Parameter:** We denote the mesh size by h, defined as,

$$h = \max_{K \in \mathcal{T}_h} \text{diam}(K)$$

  As $h \to 0$, the triangles get smaller, resulting in a fine mesh on Ω.

The Space V_h

We define V_h as a space of piecewise polynomials of degree at most k with respect to $\mathcal{F}_h$. If P_k is the space of polynomials of degree at most k, then for every $v_h \in V_h$, we require $v_h|_K \in P_k$ for every $K \in \mathcal{F}_h$. Thus V_h is finite dimensional once k is fixed and the dimension can be increased just by refining the triangulation. Thus, as $h \to 0$, we have $n(h) \to \infty$.

- **Continuity Requirements:** To ensure $V_h \subset V \subset H^1(\Omega)$, we must have $V_h \subset \mathcal{C}(\overline{\Omega})$. This requires inter-element continuity: if K_1 and K_2 are adjacent, $v_h|_{K_1}$ and $v_h|_{K_2}$ must agree on their common side.

- **Higher Order:** If $V \subset H^2(\Omega)$, we must ensure $V_h \subset \mathcal{C}^1(\overline{\Omega})$.

Basis and Sparsity

Once the triangulation and degree k are fixed, we identify **degrees of freedom** (e.g. values at certain nodes) to fix the polynomial uniquely. We associate to each node a basis function w_h^i which is unity at that specific node and zero at all others.
The dimension of the space, $\dim(V_h) = n(h)$, is the number of nodes attached to these degrees of freedom. A significant advantage of this construction is that the support of a basis function w_h^i is limited to the elements sharing that node. Consequently, the resulting stiffness matrix is sparse, as the entry $a(w_h^j, w_h^i)$ vanishes whenever the supports of the basis functions are disjoint.

4.3 Error Analysis

Céa's Lemma

Theorem 10 (Céa's Lemma)**.** *There exists a constant $C > 0$, which does not depend on h, such that*

$$\|u - u_h\| \leq C \inf_{v_h \in V_h} \|u - v_h\|. \tag{4.3}$$

Proof. Let $\alpha > 0$ be the ellipticity constant of $a(\cdot, \cdot)$ and let $M > 0$ be the constant appearing in the definition of its continuity.
Let $v_h \in V_h$ be arbitrarily chosen. By the ellipticity of the bilinear form, we have:

$$\alpha \|u - u_h\|^2 \leq a(u - u_h, u - u_h) \tag{4.4}$$

We can rewrite the second argument by adding and subtracting v_h as,

$$\alpha \|u - u_h\|^2 \leq a(u - u_h, u - v_h + v_h - u_h) = a(u - u_h, u - v_h) + a(u - u_h, v_h - u_h) \tag{4.5}$$

Now, from the definitions,

$$a(u, w_h) = \langle f, w_h \rangle \quad \forall w_h \in V_h, \tag{4.6}$$
$$a(u_h, w_h) = \langle f, w_h \rangle \quad \forall w_h \in V_h. \tag{4.7}$$

Subtracting these equations yields $a(u - u_h, w_h) = 0$ for all $w_h \in V_h$. By choosing the specific test function $w_h = v_h - u_h$, it follows that $a(u - u_h, v_h - u_h) = 0$.
Substituting this result back into (4.5), we get,

$$\alpha \|u - u_h\|^2 \leq a(u - u_h, u - v_h) \tag{4.8}$$

By applying the continuity property of $a(\cdot, \cdot)$,

$$\alpha \|u - u_h\|^2 \leq M \|u - u_h\| \|u - v_h\|,$$
$$\Rightarrow \|u - u_h\| \leq \frac{M}{\alpha} \|u - v_h\|.$$

Since this inequality holds for any $v_h \in V_h$, we take the infimum over V_h,

$$\|u - u_h\| \leq \frac{M}{\alpha} \inf_{v_h \in V_h} \|u - v_h\|. \tag{4.9}$$

This confirms (4.3) with the constant $C = M/\alpha$. □

In particular, if for $v \in V$, we can find an "interpolated element" $\Pi_h v \in V_h$ such that,

$$\|v - \Pi_h v\| \leq \phi(h), \tag{4.10}$$

where $\phi(h) \to 0$ as $h \to 0$, then by Céa's Lemma, we have

$$\|u - u_h\| \leq C\|u - \Pi_h u\| \leq C\phi(h). \tag{4.11}$$

This proves that the approximate solutions converge to the exact solution as $h \to 0$, and allows us to find a specific rate of convergence.

4.4 Example in 1-Dimensional Case

4.4.1 Problem Statement

Consider the following boundary value problem on the interval $\Omega = (0, 1)$,

$$\begin{cases} -u'' = f \text{ in } (0,1), \\ u(0) = u(1) = 0. \end{cases} \tag{4.12}$$

4.4.2 Weak Formulation

To derive the weak formulation, we multiply by a test function $v \in C_c^\infty(0,1)$ and integrate over the domain,

$$\begin{aligned} -\int_0^1 u''v\,dx &= \int_0^1 fv\,dx \\ \Rightarrow -[u'v]_0^1 + \int_0^1 u'v'\,dx &= \int_0^1 fv\,dx \end{aligned}$$

Since $v(0) = v(1) = 0$, the boundary term vanishes. The weak problem is to find $u \in H_0^1(0,1)$ such that,

$$\int_0^1 u'v'\,dx = \int_0^1 fv\,dx \quad \forall\, v \in H_0^1(0,1). \tag{4.13}$$

4.4.3 Discretization and Triangulation

Let $\Omega = (0, 1)$ be divided into n equal sub-intervals $[x_{i-1}, x_i]$ with mesh size $h = 1/n$.

- Nodes: $x_i = ih$ for $i \in \{0, 1, 2, ..., n\}$.
- Discrete Space: $V_h \subset H_0^1(0,1)$ consists of continuous piecewise linear functions.

4.4.4 Basis Functions

The basis $\{w_h^i\}_{i=1}^{n-1}$ for V_h is defined by the "hat functions":

$$w_h^i(x) = \begin{cases} \frac{1}{h}(x - x_{i-1}) & \text{if } x \in [x_{i-1}, x_i] \\ \frac{1}{h}(x_{i+1} - x) & \text{if } x \in [x_i, x_{i+1}] \\ 0 & \text{otherwise} \end{cases} \tag{4.14}$$

4.4.5 Linear System

The bilinear form $a(w_h^i, w_h^j) = \int_0^1 (w_h^i)'(w_h^j)'\,dx$ leads to a tridiagonal stiffness matrix A_h. The entries are calculated as:

- $a(w_h^i, w_h^i) = \frac{2}{h}$,
- $a(w_h^i, w_h^{i+1}) = -\frac{1}{h}$.

This results in the following system:

$$\frac{1}{h}\begin{bmatrix} 2 & -1 & 0 & \dots & 0 \\ -1 & 2 & -1 & \dots & 0 \\ 0 & -1 & 2 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & -1 \\ 0 & 0 & 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} u_h(x_1) \\ \vdots \\ \vdots \\ \vdots \\ u_h(x_{n-1}) \end{bmatrix} = \begin{bmatrix} \int_0^1 f w_h^1\,dx \\ \vdots \\ \vdots \\ \vdots \\ \int_0^1 f w_h^{n-1}\,dx \end{bmatrix} \tag{4.15}$$

The matrix above is symmetric, positive definite and sparse. Now just calculate the nodal values to get the full expression of the solution in the finite dimensional case.

Chapter 5

A Short Finite Element Implementation: Theoretical Analysis

This chapter focuses on a model problem and theoretical discussion of finite element method implementation for this. This chapter is based on the work of Dr. Carstensen, Dr. Alberty, and Dr. Funken, *"Remarks around 50 lines of Matlab: short finite element implementation. Numerical algorithms, 20(2), 117-137"*.

5.1 Finite Element Implementation: Model Problem

5.1.1 Problem Definition

Consider $\Omega \subset \mathbb{R}^2$ be a bounded Lipschitz domain with polygonal boundary Γ. Let $f \in L^2(\Omega)$, $u_D \in H^1(\Omega)$, and $g \in L^2(\Gamma_N)$. To get $u \in H^1(\Omega)$ satisfying,

$$\begin{cases} -\Delta u = f & \text{in } \Omega, \\ u = u_D & \text{on } \Gamma_D, \\ \frac{\partial u}{\partial \eta} = g & \text{on } \Gamma_N. \end{cases} \tag{5.1}$$

where $\Gamma_D \subseteq \Gamma$ has positive length and $\Gamma_N = \Gamma \setminus \Gamma_D$.

5.1.2 Solution Space

The solution u resides in the coset,

$$K = u_D + H_D^1(\Omega) = \{v \in H^1(\Omega) \mid v - u_D \in H_D^1(\Omega)\},$$

where the subspace $H_D^1(\Omega)$ is defined as,

$$H_D^1(\Omega) := \{w \in H^1(\Omega) \mid w = 0 \text{ on } \Gamma_D\}.$$

5.1.3 Weak Formulation

The weak solution $u \in K$ satisfies,

$$\int_\Omega \nabla u \cdot \nabla w \, dx = \int_\Omega f w \, dx + \int_{\Gamma_N} g w \, ds \quad \forall\, w \in H_D^1(\Omega). \tag{5.2}$$

5.1.4 Transformation to Subspace $H_D^1(\Omega)$

By defining $u = v + u_D$, we seek $v \in H_D^1(\Omega)$ such that for all $w \in H_D^1(\Omega)$,

$$\int_\Omega \nabla v \cdot \nabla w \, dx = \int_\Omega f w \, dx + \int_{\Gamma_N} g w \, ds - \int_\Omega \nabla u_D \cdot \nabla w \, dx, \quad w \in H_0^1(\Omega). \tag{5.3}$$

5.2 Galerkin Discretisation of the Problem

The above equation is discretized using the Galerkin method, where $H^1(\Omega)$ and $H_D^1(\Omega)$ are replaced by finite dimensional subspaces S and $S_D = S \cap H_D^1$, respectively. Let $U_D \in S$ be a function that approximates u_D on Γ_D. We define U_D as the nodal interpolant of u_D on Γ_D.
Then, we seek $V \in S_D$ such that

$$\int_\Omega \nabla V \cdot \nabla W \, dx = \int_\Omega f W \, dx + \int_{\Gamma_N} g W \, ds - \int_\Omega \nabla U_D \cdot \nabla W \, dx, \quad W \in S_D. \tag{5.4}$$

Let $(\eta_1, \ldots, \eta_N)$ be a basis of the finite dimensional space S, and let $(\eta_{i_1}, \ldots, \eta_{i_M})$ be a basis of S_D, where $I = \{i_1, \ldots, i_M\} \subseteq \{1, \ldots, N\}$ is an index set. Thus, (5.4) is equivalent to,

$$\int_\Omega \nabla V \cdot \nabla \eta_j \, dx = \int_\Omega f \eta_j \, dx + \int_{\Gamma_N} g \eta_j \, ds - \int_\Omega \nabla U_D \cdot \nabla \eta_j \, dx, \quad j \in I. \tag{5.5}$$

Suppose,

$$V = \sum_{k \in I} x_k \eta_k \quad \text{and} \quad U_D = \sum_{k=1}^{N} U_k \eta_k.$$

Then, we have linear system of equations,

$$Ax = b, \tag{5.6}$$

where, the coefficient matrix $A = (A_{jk})_{j,k \in I} \in \mathbb{R}^{M \times M}$ and the right-hand side $b = (b_j)_{j \in I} \in \mathbb{R}^M$ are defined as

$$A_{jk} = \int_\Omega \nabla \eta_j \cdot \nabla \eta_k \, dx \tag{5.7}$$

and

$$b_j = \int_\Omega f \eta_j \, dx + \int_{\Gamma_N} g \eta_j \, ds - \sum_{k=1}^{N} U_k \int_\Omega \nabla \eta_j \cdot \nabla \eta_k \, dx \tag{5.8}$$

The coefficient matrix is sparse, symmetric and positive definite, so (5.6) has exactly one solution $x \in \mathbb{R}^M$ which determines the solution,

$$U = U_D + V = \sum_{j=1}^{N} U_j \eta_j + \sum_{k \in I} x_k \eta_k$$

5.3 Triangulation and Basis Functions

Triangulation

Suppose the domain Ω has a polygonal boundary Γ. Then, we can say that, $\overline{\Omega} = \cup_{T\in\mathcal{T}} T$, where each T is either a closed triangle or a closed quadrilateral. Here, we follow the process of triangulation only. Following the usual process of triangulation, we have,

$$A_{jk} = \sum_{T\in\mathcal{T}} \int_T \nabla\eta_j \cdot \nabla\eta_k \, dx, \tag{5.9}$$

$$b_j = \sum_{T\in\mathcal{T}} \int_T f\eta_j \, dx + \sum_{E\subset\Gamma_N} \int_E g\eta_j \, ds - \sum_{k=1}^{N} U_k \sum_{T\in\mathcal{T}} \int_T \nabla\eta_j \cdot \nabla\eta_k \, dx. \tag{5.10}$$

Formation of Basis Functions

To implement the finite element method in $\mathbb{R}^2$, we consider the space S of continuous piecewise linear functions. For a triangular element $T \in \mathcal{T}$ with vertices $P_1(x_j, y_j)$, $P_2(x_{j+1}, y_{j+1})$, and $P_3(x_{j+2}, y_{j+2})$, we define the local basis functions $\eta_j(x, y)$ for some $j \in \mathbb{N}$ by the property:

$$\eta_j(x_k, y_k) = \delta_{jk}, \tag{5.11}$$

where,

$$\delta_{jk} = \begin{cases} 1, \ j = k \\ 0, \ j \neq k. \end{cases}$$

We consider η_j, a linear polynomial of the form $\eta_j(x, y) = a_j x + b_j y + c_j$. The coefficients are determined by the linear system:

$$\begin{pmatrix} 1 & x_j & y_j \\ 1 & x_{j+1} & y_{j+1} \\ 1 & x_{j+2} & y_{j+2} \end{pmatrix} \begin{pmatrix} c_j \\ a_j \\ b_j \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}. \tag{5.12}$$

By applying Cramer's rule, the basis function is expressed as:

$$\eta_j(x, y) = \frac{\det \begin{pmatrix} 1 & x & y \\ 1 & x_{j+1} & y_{j+1} \\ 1 & x_{j+2} & y_{j+2} \end{pmatrix}}{\det \begin{pmatrix} 1 & x_j & y_j \\ 1 & x_{j+1} & y_{j+1} \\ 1 & x_{j+2} & y_{j+2} \end{pmatrix}}. \tag{5.13}$$

5.4 Assembling the Stiffness Matrix & Right-Hand Side

The gradient $\nabla\eta_j$ is constant over T, and is given by,

$$\nabla\eta_j = \tfrac{1}{2|T|} \begin{pmatrix} y_{j+1} - y_{j+2} \\ x_{j+2} - x_{j+1} \end{pmatrix}, \text{ where } 2|T| = \det \begin{pmatrix} x_2 - x_1 & x_3 - x_1 \\ y_2 - y_1 & y_3 - y_1 \end{pmatrix}.$$

Assembling the Stiffness Matrix

The local stiffness matrix entry M_{jk} is,

$$M_{jk} = \int_T \nabla\eta_j (\nabla\eta_k)^T dx = \frac{|T|}{(2|T|)^2}(y_{j+1} - y_{j+2}, x_{j+2} - x_{j+1}) \begin{pmatrix} y_{k+1} - y_{k+2} \\ x_{k+2} - x_{k+1} \end{pmatrix}.$$

Assembling the Right-Hand Side

The integral in the right hand side is approximated by,

$$\int_T f\eta_j dx \approx \frac{1}{6} \det \begin{pmatrix} x_2 - x_1 & x_3 - x_1 \\ y_2 - y_1 & y_3 - y_1 \end{pmatrix} f(x_S, y_S).$$

For Neumann conditions on edge E with length $|E|$ is given by,

$$\int_E g\eta_j ds \approx \frac{|E|}{2} g(x_M, y_M).$$

Chapter 6

A Short Finite Element Implementation: Numerical Simulation

In this chapter, we discuss a formal mathematical and algorithmic description of a Finite Element Method (FEM) solver implemented to solve the two-dimensional Laplace equation with homogeneous Dirichlet boundary conditions on a square domain. An efficient computational framework corresponding to MATLAB simulations is provided, describing the mesh generation, stiffness matrix assembly, numerical integration and error analysis.

6.1 FEM Implementation on Square Domain

6.1.1 Model Problem

We consider the Laplace equation defined on a square domain $\Omega = (0,2) \times (0,2) \subset \mathbb{R}^2$,

$$-\Delta u(x,y) = f(x,y) \quad \text{in } \Omega, \tag{6.1}$$

subject to homogeneous Dirichlet boundary conditions on $\partial\Omega$ as,

$$u(x,y) = 0 \quad \text{on } \partial\Omega. \tag{6.2}$$

where,

$$f(x,y) = 2\left(y(2-y) + x(2-x)\right). \tag{6.3}$$

For numerical validation and testing of convergence, the analytical exact solution for this system is given by,

$$u_{\text{exact}}(x,y) = xy(2-x)(2-y). \tag{6.4}$$

6.1.2 Domain Geometry and Mesh Generation

The initial phase of the simulation consists of domain creation, mesh generation, and memory allocation. The complete, executable code is provided in Appendix. The algorithm on mesh generation is described as follows. The computational domain $\Omega = (0,2) \times (0,2)$ is created using the command,

```
    rect = [3; 4; 0; 2; 2; 0; 0; 0; 2; 2];
g = decsg(rect);
```

where, `rect` specifies a rectangle with defined x- and y-coordinates of the vertices of square region. The function `decsg` changes the geometry into a form MATLAB can use for mesh generation.
A triangular mesh is generated based on a maximum element edge-length constraint h. Now, we use,

```
    [p, e, t] = initmesh(g, 'hmax', h);
coordinates = p';
elements = t(1:3,:)';
```

where, `initmesh` initializes the mesh generation with predefined mesh size and gives three outputs, namely, the coordinates of nodes, edges, and triangles. Then, we extract the nodal coordinates, triangle data using variables, `coordinates` and `elements`, respectively.

6.1.3 Stiffness Matrix Formulation

Next, we initialize the global stiffness matrix, and the column on the right hand side. Then, using a loop for number of triangles, we update the matrix till all the elements are exhausted. Simultaneously, we update the right hand side also. The following loop works efficiently in this case.

```
for j = 1:n_elements
    nodes = elements(j,:); % Get global node indices for this triangle
    vertices = coordinates(nodes,:); % Get (x,y) coordinates for these nodes
    d = size(vertices,2);

    G = [ones(1,d+1); vertices'] \ [zeros(1,d); eye(d)];

    % Compute local 3x3 stiffness matrix
    M = det([ones(1,d+1); vertices']) * G * G' / prod(1:d);

    % Add local stiffness to global matrix (Scatter operation)
    A(nodes, nodes) = A(nodes, nodes) + M;

    % Compute local load vector contribution (Area-based)
    area = abs(det([1 1 1; vertices'])) / 2;

    % Calculate the centroid first
    centroid = sum(vertices)/3;

    % Pass the x and y components separately
    b(nodes) = b(nodes) + area * f(centroid(1), centroid(2)) / 3;
end
```

6.1.4 Dirichlet Boundary conditions and Computation at Nodes

Finally, we investigate the Dirichlet nodes, apply the boundary condition and compute the value at free nodes. We use the following MATLAB routine,

```
    u = zeros(n_nodes, 1);
bound_nodes = unique(e(1:2,:));
free_nodes = setdiff(1:n_nodes, bound_nodes);
```

```
% Set Dirichlet values (e.g., u = 0 on boundary)
u(bound_nodes) = 0;
b = b - A(:, bound_nodes) * u(bound_nodes);

% SOLVE
u(free_nodes) = A(free_nodes, free_nodes) \ b(free_nodes);
```

6.1.5 Exact Solution and Numerical Solution Plotting

In this part, we plot the numerical solution for $h = 0.5$ and exact solution. The solutions are shown as follows.

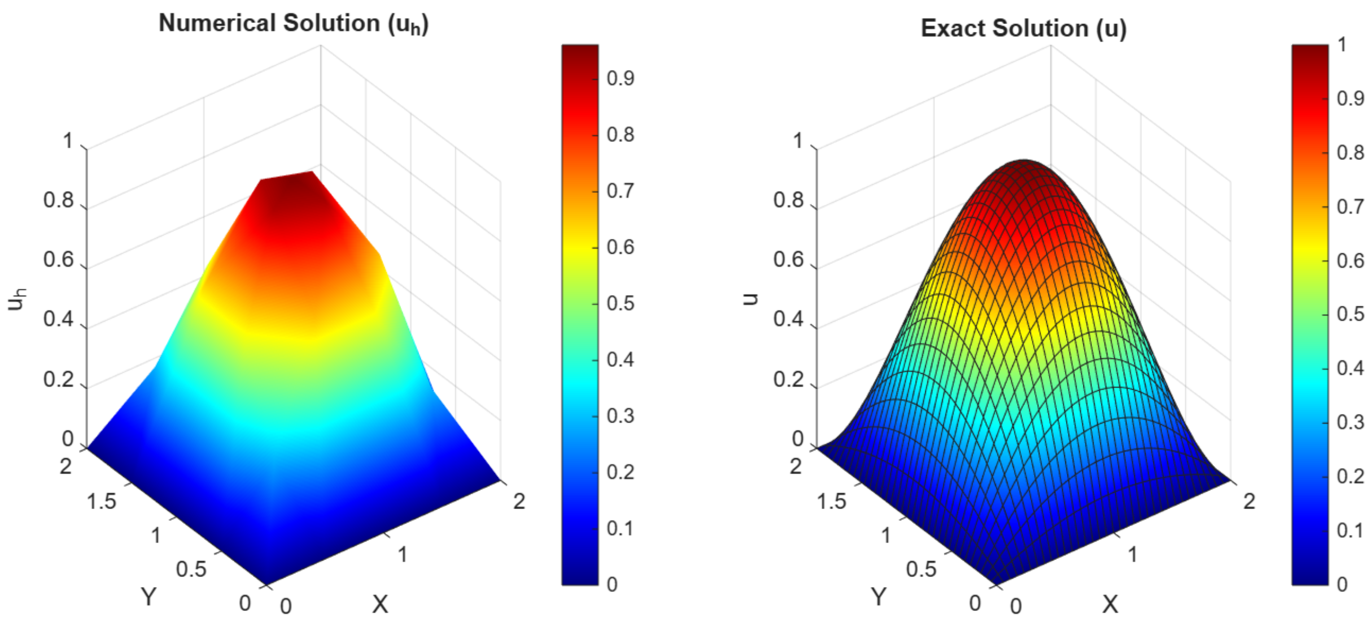

Figure 6.1: Comparison between exact solution (u) and fem induced solution (u_h) for $h = 0.5$. (Detailed MATLAB code of this plot is provided in the Appendix)

6.1.6 Error Calculation

In this part, we calculate different errors viz. maximum absolute error, L^2 error, H^1 error with respect to different values of h. Finally, we plot the Error vs. (1/h) graph to justify that as $h \to 0$, $u_h \to u$ in all the cases. The detailed table is shown in the next page.

Value of h	Maximum Absolute Error	L^2-Error	H^1-Error
$h = 0.5$	4.779771×10^{-2}	9.648578×10^{-2}	5.968972×10^{-1}
$h = 0.4$	2.581047×10^{-2}	6.234169×10^{-2}	4.842844×10^{-1}
$h = 0.3$	1.171399×10^{-2}	3.750019×10^{-2}	3.745177×10^{-1}
$h = 0.2$	6.940533×10^{-3}	1.616052×10^{-2}	2.430090×10^{-1}
$h = 0.1$	1.924976×10^{-3}	4.156899×10^{-3}	1.246075×10^{-1}
$h = 0.09$	1.632107×10^{-3}	3.356685×10^{-3}	1.118786×10^{-1}
$h = 0.08$	1.256660×10^{-3}	2.685125×10^{-3}	9.999534×10^{-2}
$h = 0.07$	1.004856×10^{-3}	2.012627×10^{-3}	8.667685×10^{-2}
$h = 0.06$	6.845399×10^{-4}	1.474371×10^{-3}	7.406463×10^{-2}
$h = 0.05$	4.811765×10^{-4}	1.004230×10^{-3}	6.111215×10^{-2}

Table 6.1: Finite Element Error Analysis on the Square Domain

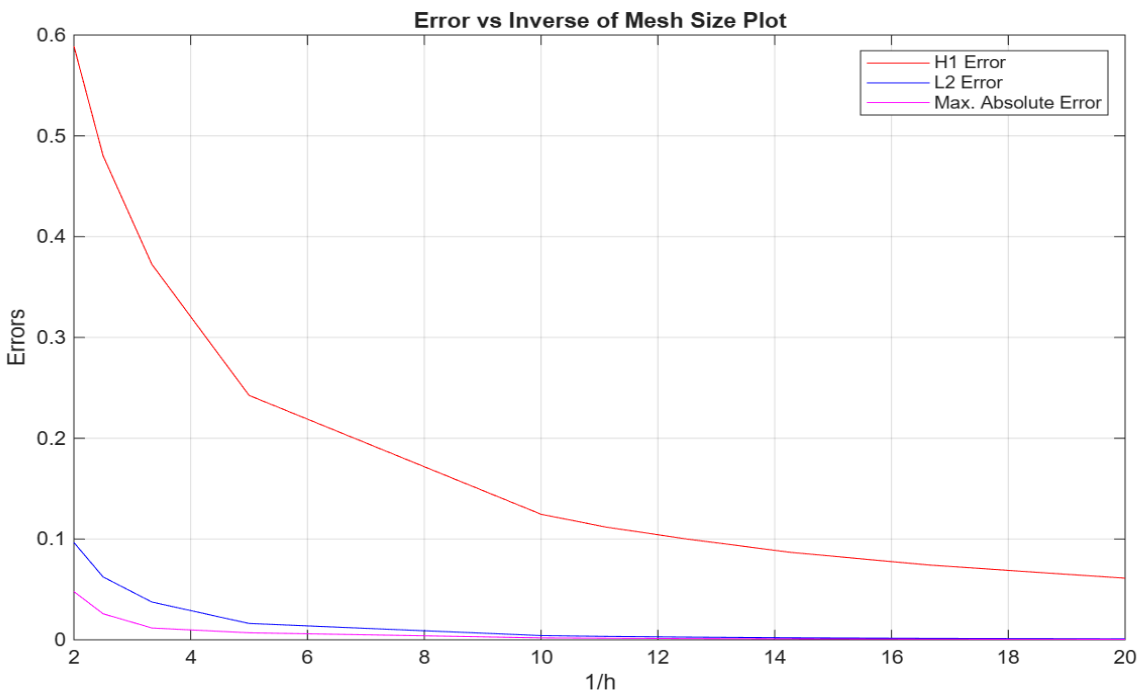

Figure 6.2: Comparison between H^1-error, L^2 error and maximum absolute error for the variation of mesh size. (Detailed MATLAB code of this comparison is provided in the Appendix)

6.2 FEM Implementation on L-Shaped Domain

6.2.1 Model Problem

We consider the Laplace equation with homogeneous Dirichlet boundary conditions. The boundary value problem is formulated as follows: Find a scalar function $u : \bar{\Omega} \to \mathbb{R}$ such that,

$$\begin{cases} -\Delta u = f & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega, \end{cases} \tag{6.5}$$

where $\Omega = (0, 2) \times (0, 2) \setminus [1, 2) \times [1, 2)$, $f : \Omega \to \mathbb{R}$ is the given function. In this setting, the point $(1, 1)$ acts as a re-entrant boundary corner, directly interacting with the mesh triangulation.

Exact Solution

To study the convergence properties of the FEM under this specific geometric configuration, we consider an exact solution $u(x, y)$ that explicitly contains a corner-like algebraic singularity at (1,1). The chosen exact solution is given by,

$$u(x, y) = \Big(x(2-x)y(2-y)(1-x)(1-y)\Big) \cdot \Big((x-1)^2 + (y-1)^2\Big)^{5/3} \tag{6.6}$$

.

The Right Hand Side

The corresponding right-hand side $f(x, y)$ is derived analytically by taking the negative Laplacian of the exact solution as follows,

$$f(x, y) = -\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right). \tag{6.7}$$

6.2.2 Exact and FEM Induced Solution Plotting

Following the similar algorithm as in square domain, we get the exact and numerical solution as follows.

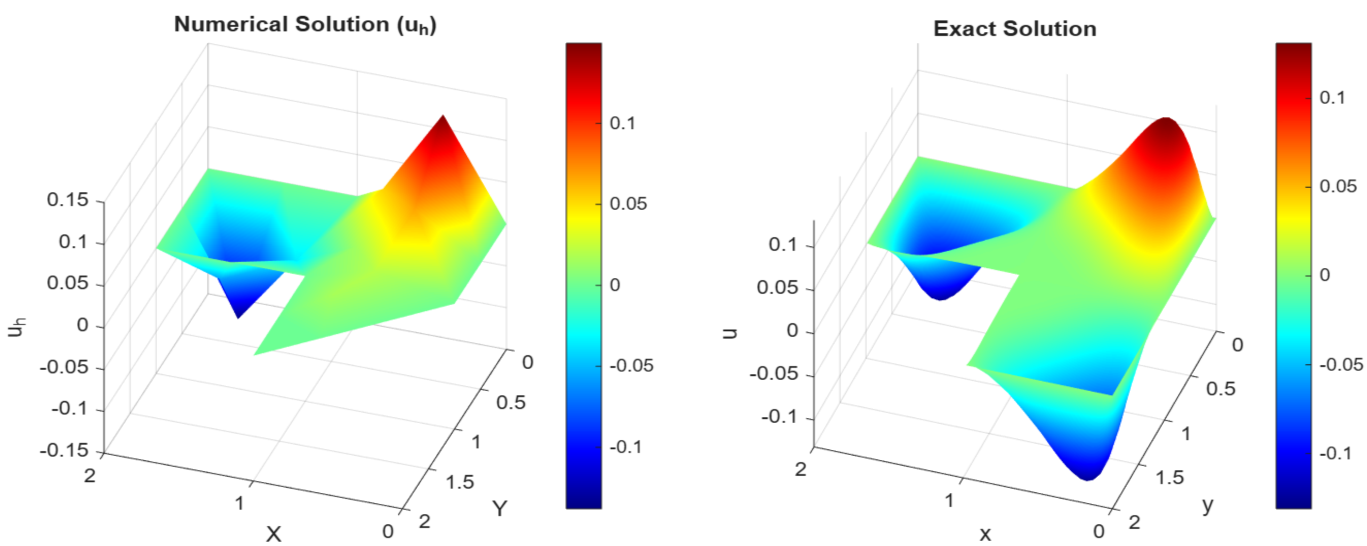

Figure 6.3: Comparison between exact solution (u) and fem induced solution (u_h) for $h = 0.5$.

6.2.3 Error Calculation

Similar to previous problem, we calculate different errors viz. maximum absolute error, L^2 error, H^1 error with respect to different values of h. Finally, we plot the Error vs. (1/h) graph to justify that as $h \to 0$, $u_h \to u$ in all the cases. The detailed table is shown in the next page.

Value of h	Maximum Absolute Error	L^2 Error	H^1-Error
$h = 0.5$	4.208357×10^{-2}	3.896254×10^{-2}	4.407125×10^{-1}
$h = 0.4$	4.429410×10^{-2}	3.023127×10^{-2}	3.374231×10^{-1}
$h = 0.3$	4.338531×10^{-2}	2.639562×10^{-2}	2.937486×10^{-1}
$h = 0.2$	4.429410×10^{-2}	1.927062×10^{-2}	2.147871×10^{-1}
$h = 0.1$	4.420508×10^{-2}	1.607015×10^{-2}	1.275054×10^{-1}
$h = 0.09$	4.429410×10^{-2}	1.589251×10^{-2}	1.141439×10^{-1}
$h = 0.08$	4.429410×10^{-2}	1.583417×10^{-2}	1.053821×10^{-1}
$h = 0.07$	4.424906×10^{-2}	1.581138×10^{-2}	9.880510×10^{-2}
$h = 0.06$	4.429410×10^{-2}	1.578339×10^{-2}	9.234671×10^{-2}
$h = 0.05$	4.427058×10^{-2}	1.576276×10^{-2}	8.438458×10^{-2}

Table 6.2: Error Analysis for the L-shaped Domain FEM Problem

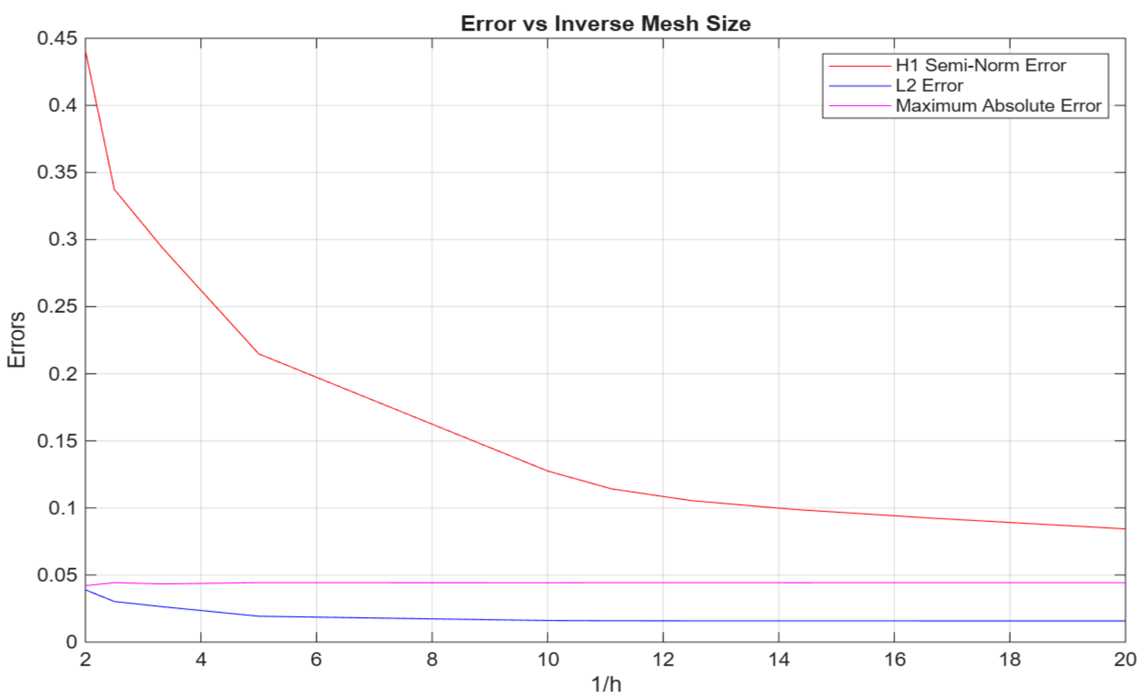

Figure 6.4: Comparison among H^1-error, L^2 error and maximum absolute error for the variation of mesh size.

6.3 Comparison of Convergence Rates for Square Domain and L-Shaped Domain

To study the convergence behaviour of the finite element approximation, we compute the Experimental Order of Convergence (EOC) using the formula,

$$\text{EOC} = \frac{\ln\left(\frac{E_{h_i}}{E_{h_{i+1}}}\right)}{\ln\left(\dfrac{h_i}{h_{i+1}}\right)}, \tag{6.8}$$

where E_{h_i} denotes the numerical error corresponding to mesh size h_i, where $i \in \mathbb{N}$. We compare the EOC for H^1-error for square and L-shaped domain in the following.

Mesh Refinement	Square Domain	L-shaped Domain
$0.50 \to 0.40$	0.91	1.20
$0.40 \to 0.30$	0.88	0.48
$0.30 \to 0.20$	1.06	0.77
$0.20 \to 0.10$	0.96	0.75
$0.10 \to 0.09$	1.06	1.05
$0.09 \to 0.08$	0.94	0.68

Table 6.3: Comparison of Experimental Order of Convergence (EOC) in the H^1-norm for the square and L-shaped domains

While the errors are decreasing for square domains according to mesh refinements, the convergence rate is much slower, sometimes fluctuating for L-shaped domain. Moreover, the maximum absolute error is seemed to almost constant after several mesh refinements. This reflects the limitation of finite element method to the domain geometry.

Chapter 7

Conclusion & Future Plan

7.1 Conclusion

In this work, the finite element method was implemented for the numerical solution of second-order elliptic equations involving Laplace operators for square and L-shaped domains with Dirichlet boundary conditions. For the square domain, where the exact solution is sufficiently smooth, the numerical schemes demonstrated the optimal convergence of the finite element analysis. The observed Experimental Order of Convergence (EOC) in the H^1-norm was approximately one. For the L-shaped domain, the presence of the re-entrant corner caused a singularity in the exact solution, resulting in reduced regularity. Consequently, the convergence rate of the finite element method was lower than the optimal rate observed in the square domain. The computed EOCs in the H^1-norm reflected this loss of regularity. The numerical experiments illustrate both the effectiveness of the finite element method and the influence of domain geometry on convergence behaviour of the numerical solutions.

7.2 Possible Future Extensions

Various extensions of this project might be considered in future investigations, as mentioned in the following:

- Implementation of adaptive mesh refinement techniques near the re-entrant corner to improve convergence rates.
- Study of higher-order finite element methods and investigation of their convergence behaviour.
- Study of a posteriori error estimators for adaptive FEM.
- Extension of the implementation to three-dimensional elliptic boundary value problems.
- Numerical treatment of more general conditions such as Neumann and mixed boundary conditions.
- Study of time-dependent partial differential equations using finite element discretization.

Chapter 8

Appendix

MATLAB Code for Numerical Solution

```
    clear;
% creating square domain [0,2]*[0,2]
rect =[3; 4; 0; 2; 2; 0; 0; 0; 2; 2];
g=decsg(rect);
%setting the mesh size
h = 0.5;
%Generating the mesh
[p,e,t]=initmesh(g, 'hmax',h);
%Extracting the data into variables
coordinates = p';
elements=t(1:3,:)';
n_nodes = size(coordinates,1);
n_elements = size(elements,1);
f=@(x,y) 2*(y.*(2-y) + x.*(2-x));
% Initialize Global Matrix and Vector
A = sparse(n_nodes, n_nodes);
b = zeros(n_nodes, 1);


%% 2. ASSEMBLY OF STIFFNESS MATRIX (A) & VOLUME FORCE (b)
for j = 1:n_elements
    nodes = elements(j,:); % Get global node indices for this triangle
    vertices = coordinates(nodes,:); % Get (x,y) coordinates for these nodes
    d=size(vertices,2);
    G=[ones(1,d+1);vertices']\[zeros(1,d);eye(d)];
    % Compute local 3x3 stiffness matrix
    M = det([ones(1,d+1);vertices'])*G*G'/prod(1:d);

    % Add local stiffness to global matrix (Scatter operation)
    A(nodes, nodes) = A(nodes, nodes) + M;
```

```matlab
    % Compute local load vector contribution (Area-based)
    area = abs(det([1 1 1; vertices'])) / 2;
    % Calculate the centroid first
    centroid = sum(vertices)/3;

    % Pass the x and y components separately
    b(nodes) = b(nodes) + area * f(centroid(1), centroid(2)) / 3;
end


%% 4. DIRICHLET BOUNDARY CONDITIONS
% Find free nodes (those not on the Dirichlet boundary)
u = zeros(n_nodes, 1);
bound_nodes = unique(e(1:2,:));
free_nodes = setdiff(1:n_nodes, bound_nodes);

% Set Dirichlet values (e.g., u = 0 on boundary)
u(bound_nodes) = 0;
b = b - A(:, bound_nodes) * u(bound_nodes);

%% 5. SOLVE
u(free_nodes) = A(free_nodes, free_nodes) \ b(free_nodes);

%% 6. DISPLAY RESULTS
fprintf('\nSolution at nodes:\n');
disp(u);
% 1. Define the function handle
u_exact = @(x,y) x.*(2-x).*y.*(2-y);
% 1. Extract coordinates and connectivity
X = coordinates(:, 1);
Y = coordinates(:, 2);
% Extract the node connectivity (columns 2, 3, and 4)
tri = elements(:, 1:3);
% 1. Create a new figure window
figure('Name', 'FEM Results Comparison', 'Position', [100, 100, 1000, 400]);

% 2. Left Subplot: Numerical Solution
subplot(1, 2, 1); % (1 row, 2 columns, 1st position)
trisurf(tri, X, Y, u);
shading interp; colormap jet; colorbar;
title('Numerical Solution (u_h)');
xlabel('X'); ylabel('Y'); zlabel('u_h');
view(3); grid on;

% 3. Right Subplot: Exact Solution
subplot(1, 2, 2); % (1 row, 2 columns, 2nd position)
% Evaluating the exact formula at the mesh nodes
```

```
fsurf(u_exact, [0 2 0 2]);
shading interp;
colormap jet; colorbar;
title('Exact Solution (u)');
xlabel('X'); ylabel('Y'); zlabel('u');
view(3);
```

MATLAB Code for Error Analysis

```
clear;
clc;

% creating square domain [0,2]*[0,2]
rect =[3; 4; 0; 2; 2; 0; 0; 0; 2; 2];
g = decsg(rect);

% setting the mesh sizes
h = [0.5 0.4 0.3 0.2 0.1 0.09 0.08 0.07 0.06 0.05];

k = 1./h;

% storage arrays
all_abs_err = zeros(size(h));
all_l2_err  = zeros(size(h));
all_h1_err  = zeros(size(h));

for i = 1:length(h)

    %% GENERATING THE MESH
    [p,e,t] = initmesh(g,'hmax',h(i));

    %% EXTRACTING DATA
    coordinates = p';
    elements = t(1:3,:)';

    n_nodes = size(coordinates,1);
    n_elements = size(elements,1);

    %% SOURCE FUNCTION
    f = @(x,y) 2*(y.*(2-y) + x.*(2-x));

    %% INITIALIZE GLOBAL MATRIX AND VECTOR
    A = sparse(n_nodes,n_nodes);
    b = zeros(n_nodes,1);
```

```
l2_error_sq = 0;
h1_error_semi_sq = 0;

%% EXACT SOLUTION
u_exact = @(x,y) x.*(2-x).*y.*(2-y);

du_dx_exact = @(x,y) (2 - 2*x) .* (2*y - y.^2);

du_dy_exact = @(x,y) (2*x - x.^2) .* (2 - 2*y);

%% ASSEMBLY OF STIFFNESS MATRIX AND LOAD VECTOR

for j = 1:n_elements

    nodes = elements(j,:);

    vertices = coordinates(nodes,:);

    d = size(vertices,2);

    G = [ones(1,d+1);vertices'] \ [zeros(1,d);eye(d)];

    %% LOCAL STIFFNESS MATRIX

    M = det([ones(1,d+1);vertices']) * G * G' / prod(1:d);

    A(nodes,nodes) = A(nodes,nodes) + M;

    %% LOCAL LOAD VECTOR

    area = abs(det([1 1 1; vertices'])) / 2;

    %% 3-POINT QUADRATURE

    mid1 = 0.5 * (vertices(1,:) + vertices(2,:));
    mid2 = 0.5 * (vertices(2,:) + vertices(3,:));
    mid3 = 0.5 * (vertices(3,:) + vertices(1,:));

    f_val1 = f(mid1(1),mid1(2));
    f_val2 = f(mid2(1),mid2(2));
    f_val3 = f(mid3(1),mid3(2));

    b_local = (area/3) * (1/3) * ...
              (f_val1 + f_val2 + f_val3) * ones(3,1);

    b(nodes) = b(nodes) + b_local;
```

```
end

%% DIRICHLET BOUNDARY CONDITIONS

u = zeros(n_nodes,1);

bound_nodes = unique(e(1:2,:));

free_nodes = setdiff(1:n_nodes,bound_nodes);

u(bound_nodes) = 0;

b = b - A(:,bound_nodes) * u(bound_nodes);

%% SOLVE

u(free_nodes) = A(free_nodes,free_nodes) \ b(free_nodes);

%% MAXIMUM ERROR

X = coordinates(:,1);
Y = coordinates(:,2);

u_exact_nodes = u_exact(X,Y);

absolute_error = max(abs(u - u_exact_nodes));

fprintf('Absolute (Max) Error: %.6e\n', absolute_error);

%% ERROR ANALYSIS

for j = 1:n_elements

    nodes = elements(j,:);

    vertices = coordinates(nodes,:);

    area = abs(det([1 1 1; vertices'])) / 2;

    d = 2;

    G = [ones(1,d+1); vertices'] \ [zeros(1,d); eye(d)];

    grad_uh = u(nodes)' * G;

    %% MIDPOINTS
```

```
mid1 = 0.5 * (vertices(1,:) + vertices(2,:));
mid2 = 0.5 * (vertices(2,:) + vertices(3,:));
mid3 = 0.5 * (vertices(3,:) + vertices(1,:));

%% EXACT SOLUTION

u_ex_m1 = u_exact(mid1(1),mid1(2));
u_ex_m2 = u_exact(mid2(1),mid2(2));
u_ex_m3 = u_exact(mid3(1),mid3(2));

%% NUMERICAL SOLUTION

u_h_m1 = 0.5 * (u(nodes(1)) + u(nodes(2)));
u_h_m2 = 0.5 * (u(nodes(2)) + u(nodes(3)));
u_h_m3 = 0.5 * (u(nodes(3)) + u(nodes(1)));

%% EXACT GRADIENTS

grad_ex_m1 = [du_dx_exact(mid1(1),mid1(2)), ...
              du_dy_exact(mid1(1),mid1(2))];

grad_ex_m2 = [du_dx_exact(mid2(1),mid2(2)), ...
              du_dy_exact(mid2(1),mid2(2))];

grad_ex_m3 = [du_dx_exact(mid3(1),mid3(2)), ...
              du_dy_exact(mid3(1),mid3(2))];

%% L2 ERROR

l2_err_m1 = (u_h_m1 - u_ex_m1)^2;
l2_err_m2 = (u_h_m2 - u_ex_m2)^2;
l2_err_m3 = (u_h_m3 - u_ex_m3)^2;

l2_error_sq = l2_error_sq + ...
    area * (1/3) * ...
    (l2_err_m1 + l2_err_m2 + l2_err_m3);

%% H1 SEMI-NORM ERROR

diff_grad_m1 = grad_uh - grad_ex_m1;
diff_grad_m2 = grad_uh - grad_ex_m2;
diff_grad_m3 = grad_uh - grad_ex_m3;

h1_err_m1 = diff_grad_m1(1)^2 + diff_grad_m1(2)^2;
h1_err_m2 = diff_grad_m2(1)^2 + diff_grad_m2(2)^2;
h1_err_m3 = diff_grad_m3(1)^2 + diff_grad_m3(2)^2;
```

```
            h1_error_semi_sq = h1_error_semi_sq + ...
                area * (1/3) * ...
                (h1_err_m1 + h1_err_m2 + h1_err_m3);

        end

        %% FINAL ERRORS

        L2_error = sqrt(l2_error_sq);

        H1_error = sqrt(h1_error_semi_sq);

        %% STORE ERRORS

        all_abs_err(i) = absolute_error;
        all_l2_err(i)  = L2_error;
        all_h1_err(i)  = H1_error;

        %% DISPLAY

        fprintf('L2 Error: %.6e\n', L2_error);
        fprintf('H1 Semi-Norm Error: %.6e\n\n', H1_error);

    end

    %% PLOTS

    figure;

    plot(k, all_h1_err,'r');
    hold on;

    plot(k, all_l2_err,'b');

    plot(k, all_abs_err,'m');

    grid on;
    legend('H1 Error','L2 Error','Max. Absolute Error');
    title('Error vs Inverse of Mesh Size Plot');
    xlabel('1/h');
    ylabel('Errors');
```

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